

BURP SUITE PROJECT REPORT

REQUEST CAPTURING AND FORWARDING

Detailed Project Report – Authorized Academic Lab Environment

Document Type	Web Application Security / Practical Project Report
Team Name	Academic Beleivers
Tool	Burp Suite Community Edition
Application	Gin and Juice Shop
Project Name	Request Capturing and Forwarding
Environment	Authorized Academic Lab
Prepared By	Academic Beleivers Team
Classification	Academic / Authorized Lab Use

Team Members

Poorvika • Thejesh Kumar • Niluffur • Rakshitha • Dhanushree • Srinivasalu

Project Focus

This report documents the practical Burp Suite workflow used by the team: enabling Intercept, capturing browser requests, observing them, forwarding the requests, and observing the resulting application page.

1. Work Distribution

The project responsibilities were divided among all six team members. Each member was assigned a specific area so that the work covered the project from preparation and installation through practical testing and analysis.

Team Member	Assigned Work & Contribution
Thejesh Kumar	Handled the project background and information collection. Collected the basic concepts needed to understand Burp Suite, proxy-based testing, and the purpose of request interception.
Srinivasalu (Srinu)	Installed Burp Suite Community Edition successfully and supported the initial setup. Ensured the tool was ready for the practical browser-based activity.
Dhanushree (Dhanu)	Performed intercept and capture testing. Focused on keeping Intercept ON and observing requests generated when different actions were performed in the application.
Niluffur	Worked on forwarding captured requests and observing the resulting application behavior. Focused on understanding what happened after selecting Forward and connecting it with the response.
Rakshitha	Focused on analyzing the requests captured by Burp Suite. Observed the request information and related it to the browser action that generated the traffic.
Poorvika	Studied the basic concepts of requests and responses. Helped the team understand how a browser request travels through Burp Suite and how the application response appears after forwarding.

Team Contribution

Together, the assigned responsibilities covered project background, installation, interception, request analysis, forwarding, response observation, and request/response study.

Academic Beleivers – Team Project

2. Project Introduction

Burp Suite is used as an interception proxy for observing communication between a browser and a web application. In this academic project, the team used Burp Suite Community Edition with its browser to access the Gin and Juice Shop application.

The main project activity was request capturing and forwarding. Intercept was kept ON so that requests generated by browser actions could be paused in Burp Suite. The team then observed the captured request and selected Forward to allow it to continue to the application.

When Products or another product-related option was selected, the browser could appear to stop until the captured request was forwarded. After forwarding, the requested page opened in the browser. The team repeated this process with different application actions to understand the workflow clearly.

Project Workflow

Browser Action → HTTP Request → Burp Suite Intercept → Request Observation → Forward → Application → Response / Browser Page

Project Scope

The documented activity is limited to authorized academic lab work involving request interception, observation, forwarding, and application/response observation. It is not presented as a complete vulnerability assessment.

3. Project Objectives

The project was designed to give the team a practical understanding of how Burp Suite intercepts and releases normal web traffic in an authorized lab environment.

The objectives were:

- Understand the role of Burp Suite as an interception proxy.
- Use the Burp Suite browser to access the lab application.
- Enable Intercept and capture browser-generated requests.
- Observe basic request information in Burp Suite.
- Use Forward to release an intercepted request.
- Observe the corresponding page after forwarding.
- Develop a basic understanding of HTTP requests and responses.

Objective of the Practical Demonstration

The practical demonstration was centered on one simple cycle: perform an action in the browser, allow Burp Suite to capture the generated request, inspect the request, select Forward, and observe the resulting page.

Scope

Only the request capturing, forwarding, and related response/application behavior described by the team is documented.

4. Tools, Environment & Methodology

Tool / Component	Purpose in the Project
Burp Suite Community Edition	Intercepted and allowed browser requests to be observed and forwarded.
Burp Suite Browser	Used to open and interact with the lab application.
Gin and Juice Shop	Application used to generate normal browser requests for the exercise.

Methodology

Preparation → Open Burp Suite → Open Burp Browser → Open the application → Turn Intercept ON → Perform a browser action → Capture and inspect request → Click Forward → Observe the resulting page → Repeat with other actions → Document evidence.

Central Workflow

Browser Action → HTTP Request → Burp Suite Intercept → Request Inspection → Forward → Application → Response / Browser Page

5. Installation & Initial Setup

The first stage of the project was to install and prepare Burp Suite Community Edition. Srinivasalu (Srinu) was assigned the installation responsibility and successfully completed the setup.

After launching Burp Suite, the team used the built-in browser to access the Gin and Juice Shop application. The browser was used for normal navigation so that requests could be generated naturally and observed through Burp Suite.

The team then prepared for interception by opening the Proxy / Intercept area and switching Intercept ON. This was the key setting required for the request-capturing activity.

Setup Sequence

1. Install Burp Suite Community Edition.
2. Launch Burp Suite.
3. Open the built-in browser.
4. Enter the lab application's address.
5. Enable Intercept.
6. Perform a normal application action to generate a request.

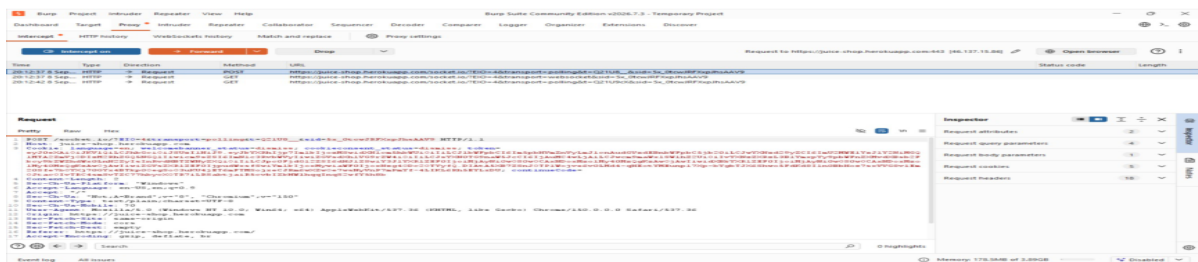


Figure 1: Burp Suite / application evidence from the team's supplied screenshots.

6. Intercept & Request Capturing

The practical demonstration started with Intercept enabled. The team opened the application in the Burp Suite browser and performed normal navigation.

When a request was generated, Burp Suite captured it and held it for inspection. The captured request could be viewed in the Proxy / Intercept interface instead of being immediately delivered to the application.

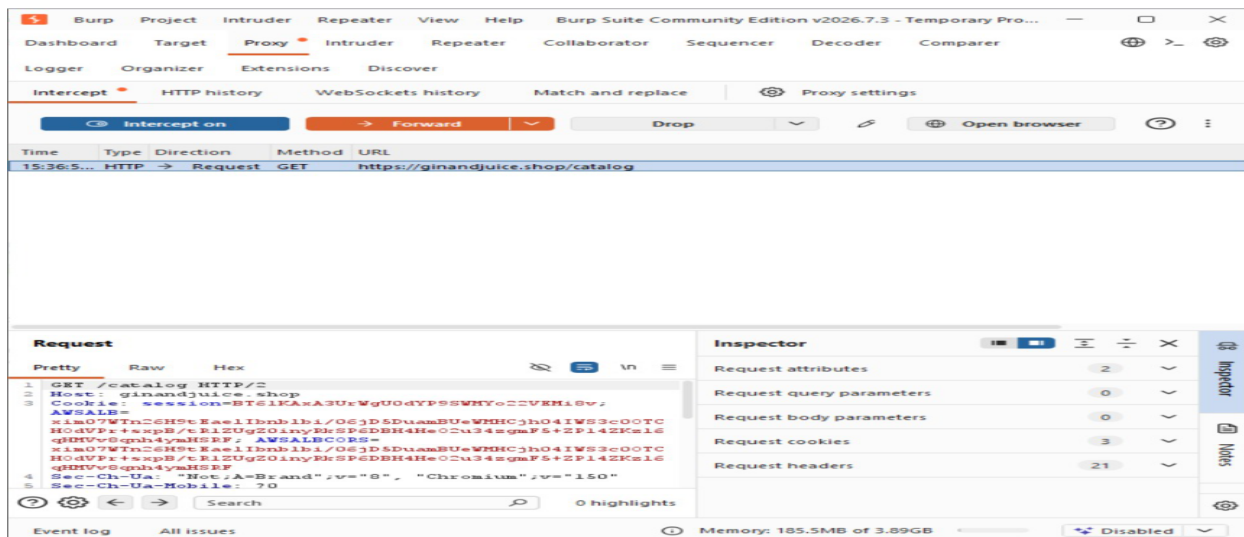


Figure 2: Captured request shown in Burp Suite during the practical activity.

Observation

The browser could appear to wait because the request was being held by Intercept. This was expected behavior for the demonstration.

7. Products Request & Forwarding

The team clicked Products in the application. Because Intercept was enabled, the browser-generated request was captured by Burp Suite.

The Products page did not immediately proceed while the request was waiting for action in Burp Suite. The team then selected Forward, allowing the request to continue.

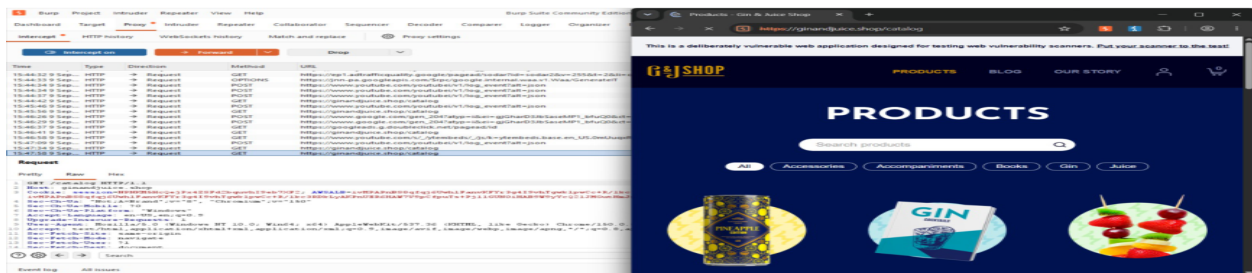


Figure 3: Products page visible after the request was allowed to continue.

Practical Result

Intercept paused the request. Forward released it. The browser then proceeded to the requested Products content.

8. Repeated Activities & Application Navigation

The same basic process was repeated with other application interactions. The purpose of repeating the activity was to confirm that different browser actions generate requests that can be observed through Burp Suite.

The team selected product categories and other product-related options. Burp Suite continued to show the resulting browser traffic, allowing the team to connect the visible application action with the corresponding request.

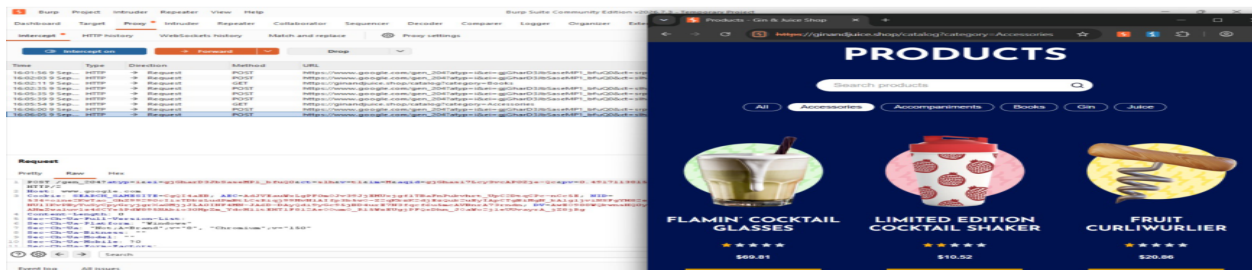


Figure 4: Representative category/product interaction from the supplied screenshots.

Duplicate screenshots from the original evidence set were intentionally omitted. Only representative images that add evidence to the explanation are included.

9. Further Request Observation

The team also performed additional navigation in the application to observe how Burp Suite records traffic for different pages. This reinforced the main workflow without changing the scope of the project.

The screenshot below shows another application navigation event together with Burp Suite request history. It demonstrates that browser navigation and the corresponding web traffic can be viewed together during the lab activity.

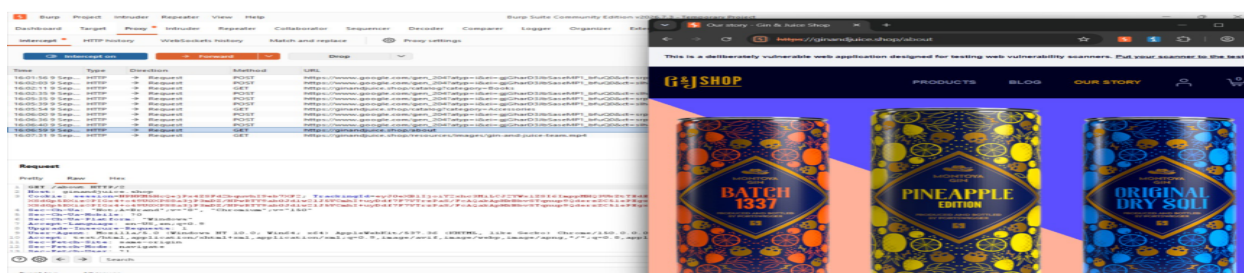


Figure 5: Representative additional application navigation captured alongside Burp Suite traffic.

Evidence Selection

Five representative screenshots were selected from the supplied PDF because several other screenshots were duplicates or showed substantially the same state.

10. Request / Response Analysis

Request Analysis

A web request is generated when the browser asks the application for a resource or performs an action. During this project, the team observed captured requests in Burp Suite and related them to the browser actions that produced them.

Response Observation

After an intercepted request was forwarded, the application could process it and the browser could display the requested content. This gave the team a practical way to understand the connection between the request and the resulting application behavior.

Member Contributions to Analysis

Niluffur: Focused on forwarding captured requests and observing the resulting application behavior.

Rakshitha: Focused on analyzing captured requests and relating them to browser actions.

Poorvika: Focused on the basic concepts of requests and responses.

Key Understanding

The exercise showed the team that browser actions create requests, Intercept can temporarily hold those requests, and Forward allows them to continue so the application can respond.

11. Results, Limitations & Conclusion

Results

The team successfully demonstrated the basic Intercept and Forward workflow. Products and other application navigation actions produced requests that could be captured, observed, and then released using Forward.

Limitations

The project was limited to request capturing, observation, forwarding, and application behavior. It did not include a full vulnerability assessment, exploitation, or claims of specific security vulnerabilities. The supplied screenshots also contained duplicate views, so only five representative screenshots were retained.

Conclusion

This project provided the Academic Beleivers team with a practical introduction to Burp Suite as an interception proxy. By enabling Intercept, generating requests through the Gin and Juice Shop application, observing the captured traffic, and selecting Forward, the team gained a clear understanding of how web requests can be paused and then allowed to continue. The activity also strengthened the team's basic understanding of web requests, responses, and browser-to-application communication.

Final Status

Completed as an academic / authorized lab exercise focused on Burp Suite request capturing and forwarding.

Academic Beleivers / Academic / Authorized Lab Use